

# Huong dan bat dau test routes Swagger

Parking Building Management System Backend - cap nhat 2026-06-15

Base URL mac dinh: <http://localhost:5000>. Swagger UI: <http://localhost:5000/api-docs>.

## 1. Chuan bi truoc khi vao Swagger

Step: Chay SQL. Neu tao DB moi, chay db/schema.sql. Neu DB da co, chay db/migration\_20260612\_mvp\_completion.sql mot lan.

Step: Seed du lieu demo: npm run seed:users, npm run seed:floors, npm run seed:vehicles.

Step: Start server: npm run dev. Kiem tra GET /api/health truoc.

Step: Mo /api-docs, bam Authorize va dan token theo tung role dang test.

**NOTE: Tai khoan seed: admin@test.com, manager@test.com, staff@test.com, user1@test.com. Mat khau mac dinh: 123456.**

## 2. Thu tu lay token

Step: Login ADMIN: POST /api/auth/login voi admin@test.com. Dung token nay de duyet user/vehicle.

Step: Login MANAGER: manager@test.com. Dung token nay de cau hinh toa nha, tang, slot, gia, goi thang, bao cao.

Step: Login STAFF: staff@test.com. Dung token nay de check-in, check-out, validate QR, ghi vi pham.

Step: Login USER: user1@test.com. Dung token nay de dang ky xe, mua goi, xem QR cua minh.

**NOTE: User moi dang ky bang /api/auth/register se co status PENDING. Admin can cap nhat sang ACTIVE truoc khi user login.**

## 3. Test theo dung luong nghiep vu MVP

### A. Health, Auth, Account

Step: GET /api/health de chac DB ket noi.

Step: POST /api/auth/register tao user moi neu can.

Step: ADMIN: GET /api/admin/users?status=PENDING.

Step: ADMIN: PATCH /api/admin/users/{id}/role-status body { role: 'USER', status: 'ACTIVE' } de duyet tai khoan.

### B. Building, Floor, Slot

Step: MANAGER: GET /api/buildings. Neu chua co, POST /api/buildings.

Step: MANAGER: POST /api/floors tao tang MOTORBIKE voi capacity, khong tao slot le.

Step: MANAGER: POST /api/floors tao tang CAR voi slotCount hoac danh sach slots.

Step: MANAGER: GET /api/floors?buildingId=1 de xem tang va slot da sinh.

Step: MANAGER: GET /api/slots/floor/{floorId}, PUT/PATCH slot neu can doi AVAILABLE/MAINTENANCE/LOCKED.

### C. Gia va goi thang

Step: MANAGER: POST /api/pricing-policies tao gia MOTORBIKE/TURN amount 4000.

Step: MANAGER: POST /api/pricing-policies tao gia CAR/HOURLY amount 20000.

Step: MANAGER: POST /api/package-plans tao goi MOTORBIKE 30 ngay.

Step: MANAGER: POST /api/package-plans tao goi CAR 30 ngay. Goi CAR se dung khi dang ky slot oto.

## D. The QR tam

Step: MANAGER: POST /api/temp-qr-cards với cardCode TEMP-001, status READY.

Step: STAFF: GET /api/temp-qr-cards?status=READY để chọn thẻ cho khách vắng lại/gửi theo lượt.

## E. Vehicle và duyệt xe

Step: USER: POST /api/vehicles đăng ký xe MOTORBIKE hoặc CAR.

Step: ADMIN: GET /api/vehicles/admin?status=PENDING nếu route này đang có trong Swagger.

Step: ADMIN: approve/reject vehicle theo route Vehicles Admin trong Swagger.

## F. Mua gói/QR pass

Step: Xe máy có thể mua gói: USER POST /api/package-plans/{id}/buy body { vehicleId, bankCode: 'NCB', locale: 'vn' }.

Step: Sau khi có paymentUrl, test VNPay sandbox hoặc gọi /api/payments/vnpay-return với query VNPay hợp lệ.

Step: Ô tô phải đăng ký slot: USER POST /api/slot-registrations body { vehicleId, slotId, packagePlanId, bankCode: 'NCB' }.

Step: Khi payment SUCCESS, backend tự tạo QR pass. USER xem bằng GET /api/qr-passes/my.

## G. Check-in

Step: Thẻ thang hợp lệ: STAFF POST /api/qr-passes/validate body { qrCode }.

Step: Thẻ thang xe máy: STAFF POST /api/parking-sessions/check-in body { qrCode, plateNumber, vehicleType: 'MOTORBIKE', buildingId }.

Step: Thẻ thang ô tô: STAFF POST /api/parking-sessions/check-in body { qrCode, plateNumber, vehicleType: 'CAR', slotId }.

Step: Gửi theo lượt/khách vắng lại xe máy: STAFF check-in body { plateNumber, vehicleType: 'MOTORBIKE', buildingId, tempQrCardCode: 'TEMP-001' }.

Step: Gửi theo giờ/khách vắng lại ô tô: STAFF check-in body { plateNumber, vehicleType: 'CAR', slotId, tempQrCardCode: 'TEMP-001' }.

**NOTE: Nếu không có thẻ thang, backend bắt buộc có tempQrCardId hoặc tempQrCardCode.**

## H. Vi phạm và phạt

Step: STAFF: POST /api/violations body { parkingSessionId, violationType: 'WRONG\_SLOT', penaltyFee: 50000, note }.

Step: STAFF: GET /api/violations?parkingSessionId={id} để kiểm tra vi phạm đang OPEN/RESOLVED.

**NOTE: Khi check-out, backend tự cộng các vi phạm OPEN/RESOLVED vào tổng tiền. Thẻ thang vẫn phải trả phí phạt.**

## I. Check-out và thanh toán

Step: STAFF: POST /api/parking-sessions/{id}/check-out. Nếu có tiền, truyền paymentMethod CASH, CARD hoặc VNPAY.

Step: STAFF: POST /api/parking-sessions/check-out-by-qr body { qrCode, paymentMethod: 'CASH' } nếu muốn quét QR lúc ra.

Step: Xe máy vắng lại tính 4000/lượt nếu pricing policy không bị đổi.

Step: Ô tô vắng lại tính 20000/giờ, làm tròn lên tối thiểu 1 giờ.

Step: VNPay checkout tạo paymentUrl và session sang PENDING\_PAYMENT. Callback SUCCESS mỗi dòng session và trả slot/capacity.

## J. Báo cáo

Step: MANAGER: GET /api/reports/traffic.

Step: MANAGER: GET /api/reports/motorbike-capacity.  
Step: MANAGER: GET /api/reports/car-slots.  
Step: MANAGER: GET /api/reports/revenue.  
Step: MANAGER: GET /api/reports/qr-passes.  
Step: MANAGER: GET /api/reports/violations.

## 4. Payload mau nhanh

```
Pricing xe may: { "vehicleType": "MOTORBIKE", "pricingType": "TURN", "amount": 4000 }  
Pricing oto: { "vehicleType": "CAR", "pricingType": "HOURLY", "amount": 20000 }  
Temp QR: { "cardCode": "TEMP-001", "status": "READY" }  
Goi xe may: { "name": "Goi xe may 30 ngay", "vehicleType": "MOTORBIKE", "price": 120000, "durationDays": 30 }  
Goi oto: { "name": "Goi oto 30 ngay", "vehicleType": "CAR", "price": 1500000, "durationDays": 30 }  
Violation: { "parkingSessionId": 1, "violationType": "WRONG_SLOT", "penaltyFee": 50000 }
```

## 5. Loi hay gap

- 403: dang dung sai token role, hoac user moi van PENDING.
- Check-in the thang bao loi qrCode: can tao/lay QR pass bang /api/qr-passes/my hoac validate truoc.
- Check-in luot bao loi temp QR: the QR tam phai ton tai va status READY.
- Slot oto khong vao duoc: slot phai AVAILABLE, rieng xe co slot thang co the vao slot RESERVED cua minh.
- VNPay PENDING: session/registration chi hoan tat khi /api/payments/vnpay-return hoac /api/payments/vnpay-ipn SUCCESS.
- Migration bao duplicate column: DB da co mot phan thay doi; bo cac dong ALTER da co san hoac tao DB fresh bang schema.sql.

## 6. Checklist ket thuc

- Admin duyet account va vehicle.
- Manager tao floor MOTORBIKE theo capacity, floor CAR theo slot.
- Manager tao pricing policy va package plan.
- Staff co the QR tam READY.
- User co QR pass cho monthly/slot registration sau thanh toan.
- Staff check-in, ghi vi pham, check-out va thu phi thanh cong.
- Bao cao hien so lieu thay doi sau cac luong test.